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1. Remember that Σ is the collection of all well-formed states, and $\Sigma_{\perp} = \Sigma \cup \{\perp\}$. Decide true or false for each of the following statements, justify your answers briefly.
 - a. If $\sigma(p) = \perp$, then $\not\models p$.
 - b. Let $\tau \in \Sigma_{\perp}$, then $\tau \models p$ or $\tau \models \neg p$.
 - c. Let $\Sigma_0 \subseteq \Sigma$ and $\Sigma_0 \models p$, also let $\tau \models p$; then $\Sigma_0 \cup \{\tau\} \models p$.
 - d. $\emptyset \models p$ and $\emptyset \models \neg p$ ($\emptyset$ represents an empty collection of states).
 - e. Let $\Sigma_0 \subset \Sigma$, then $\Sigma_0 \models x > 1$ or $\Sigma_0 \models x \leq 1$.
2. Calculate denotational semantics for the following nondeterministic programs.
 - a. Let $IF \equiv \text{if } x > y \rightarrow x := x - 1 \sqcap x > y \rightarrow y := y + 1 \sqcap x + y = 4 \rightarrow x := y/x \sqcap x + y = 4 \rightarrow x := x/y \text{ fi}$, and let $\sigma_1 = \{x = 3, y = 1\}$. Calculate $M(IF, \sigma_1)$ and show your work.
 - b. Let $IF \equiv \text{if } x > y \rightarrow x := x - 1 \sqcap x > y \rightarrow y := y + 1 \sqcap x + y = 4 \rightarrow x := y/x \sqcap x + y = 4 \rightarrow x := x/y \text{ fi}$, and let $\sigma_2(x) = \sigma_2(y) = 1$. Calculate $M(IF, \sigma_2)$ and show your work.
 - c. Let $DO \equiv \text{do } x > y \rightarrow x := x - 1 \sqcap x > y \rightarrow y := y + 1 \sqcap x + y = 4 \rightarrow x := y/x \sqcap x + y = 4 \rightarrow x := x/y \text{ od}$, and let $\sigma_1 = \{x = 3, y = 1\}$. Calculate $M(DO, \sigma_1)$ and show your work.
 - d. Let $DO \equiv \text{do } x > y \rightarrow x := x - 1 \sqcap x > y \rightarrow y := y + 1 \sqcap x + y = 4 \rightarrow x := y/x \sqcap x + y = 4 \rightarrow x := x/y \text{ od}$, and let $\sigma_2(x) = \sigma_2(y) = 1$. Calculate $M(DO, \sigma_2)$ and show your work.
3. Let b be an array of size $n \geq 1$, and $\forall 0 \leq i < n. b[i] = 0 \vee b[i] = 1$. Decide which number (0 or 1) is the majority in b without counting their quantities.

Write a program named *MAJORITY* in our language that can solve the above problem and assign the majority among 0 and 1 to variable *major*. You can assume that b is written in the memory state; and to simplify the question, we artificially define $b[n] = 100$, so you don't need to worry about a possible runtime error when the array index reaches n . Your program doesn't have to be deterministic. Be careful of the grammar in our programming language.

Here are some hints:

- 1) We can use the following linear-search-like algorithm: scan the array b to pair up each 0 with a 1. Once we have some 1's left over, then 1 is the majority; once we have some 0's left over, then 0 is the majority; if we can pair up all numbers, then either of them can be the majority.
- 2) A student named Jason wrote a partial solution that could be useful to solve the above problem:

$$J \equiv \text{do } b[k_0] = 1 \rightarrow k_0 := k_0 + 1 \sqcap b[k_1] = 0 \rightarrow k_1 := k_1 + 1 \text{ od}$$

Consider k_0 and k_1 as pointers for number 0 and 1 respectively. What program J does is to find the next k_0 and k_1 such that $b[k_0] = 0$ and $b[k_1] = 1$. You can use J inside of your program.

4. Decide true or false for each of the following statements, justify your answers briefly.
 - a. If $M(S, \sigma)$ contains exactly one state, then S is a deterministic statement.
 - b. If $\sigma \not\models \{p\} S \{q\}$, then $\sigma \models p$.
 - c. If $\sigma \models_{tot} \{p\} S \{q\}$, then $\sigma \models p$.
 - d. If $\sigma \models \{p\} S \{q\}$, then $M(S, \sigma) \models q$.
 - e. If $\sigma \not\models_{tot} \{p\} S \{q\}$, then $\sigma \not\models \{p\} S \{q\}$.
5. Answer the following questions about possible values of variable x in a state. Justify your answer briefly.
 - a. Let $\perp_e \notin M(S, \sigma)$, where $S \equiv x := \text{sqrt}(x) / b[x]$ and $\sigma(b) = (3, 0, -2, 4)$. What are the possible values of $\sigma(x)$?
 - b. Let $\sigma \models \{\text{sqrt}(x) \neq 1\} x := 1/x \{F\}$. What are the possible values of $\sigma(x)$?
 - c. Let $\sigma \models \{x \neq 0\} \text{ while } x \neq 0 \text{ do } x := x - 2 \text{ od } \{x < 0\}$, what are the possible values of $\sigma(x)$?
 - d. Let $\sigma \models_{tot} \{x \neq 0\} \text{ while } x \neq 0 \text{ do } x := x - 2 \text{ od } \{x < 0\}$, what are the possible values of $\sigma(x)$?
6. Given that $M(S, \sigma) = \{\perp_e, \{x = 2\}\}$ and $\sigma \models p$; decide true or false for each of the following statements.
 - a. S is a nondeterministic statement.
 - b. $\models_{tot} \{T\} S \{q\}$
 - c. $\sigma \models \{\neg p\} S \{q\}$
 - d. $\models_{tot} \{p\} S \{F\}$
 - e. $\models \{p\} S \{F\}$
7. Let predicate function $P(k, s) \equiv s^2 \leq k \leq (s + 1)^2$. For each of the following triples, decide whether it is valid under total correctness, justify your answer briefly.
 - a. $\{P(k, s + 1)\} s := s + 1 \{P(k, s)\}$
 - b. $\{P(k, s)\} s := s + 1 \{P(k, s + 1)\}$
 - c. $\{P(k, s) \wedge s < 0\} s := s + 1; k := k + 1 \{P(k, s)\}$
 - d. $\{P(k, s) \wedge s = x\} s := s + 1 \{P(k, x)\}$
 - e. $\{P(k + 1, s + 1)\} s := s + 1; k := k + 1 \{P(k, s)\}$
8. Let $\sigma \models_{tot} \{p_1\} S \{q_1\}$ and $\sigma \models_{tot} \{p_2\} S \{q_2\}$. Decide whether σ necessarily satisfies the following triples under total correctness, justify your answer briefly.
 - a. $\{p_1 \wedge p_2\} S \{q_1 \wedge q_2\}$
 - b. $\{p_1 \vee p_2\} S \{q_1 \vee q_2\}$
 - c. $\{p_2\} S \{q_1 \rightarrow q_2\}$

Hints for questions 8:

- 1) Remember that $\sigma \models p \wedge q$ means " $\sigma \models p$ and $\sigma \models q$ "; $\sigma \models p \vee q$ means " $\sigma \models p$ or $\sigma \models q$ ".
- 2) To prove $\sigma \models_{tot} \{p\} S \{q\}$, you can prove that "if $\sigma \models p$, then $M(S, \sigma) \models q$ ".